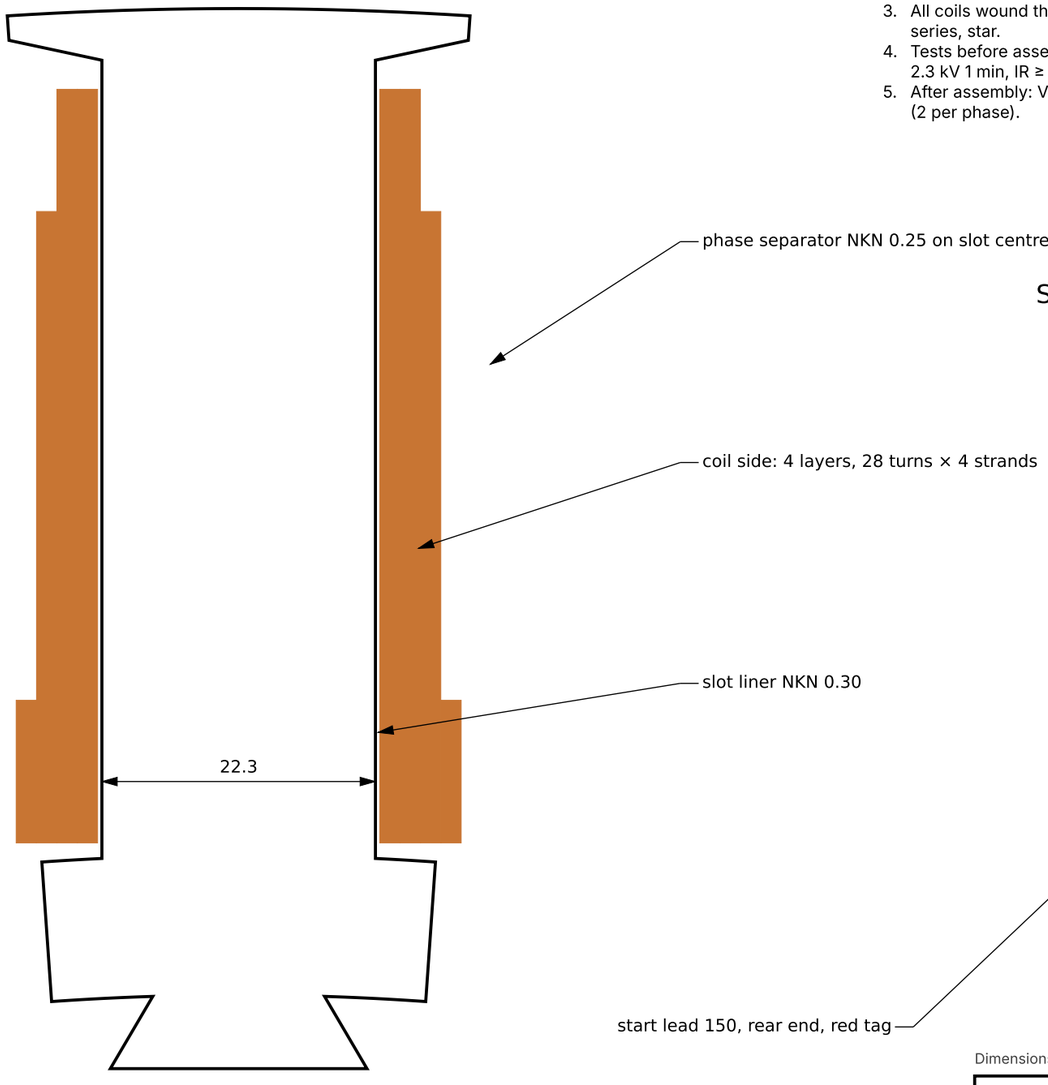
COIL SIDES IN THE SLOT (2:1)

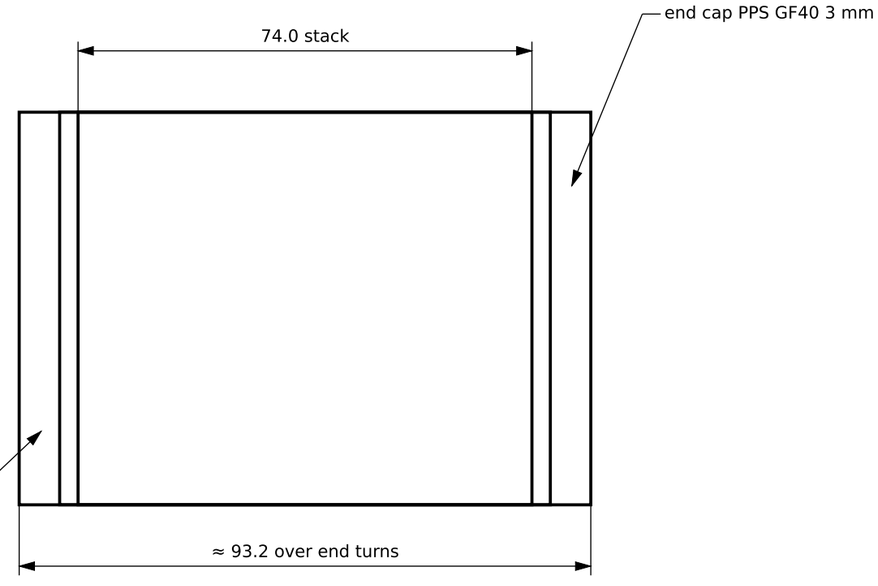


Layer (from tooth flank)	Positions	Used
1	37	37
2	37	37
3	31	31
4	17	7
5	3	0 (reserve)
Total (28 turns × 4 strands)	125	112

NOTES

1. Wind on chuck NP1-T04 by the dovetail, 4 parallel strands Ø1.55 (≤ 1.66 over enamel), 25 to 30 N per strand, layer by layer from the tooth flank, ≤ 60 rpm (doc 18 §5.6).
2. 28 turns per coil (Rev C keeps doc 18): 373 V line at 178 rpm, magnets 100 °C; 389 V at 60 °C. Wire per coil ≈ 6.6 m per strand; R20 per phase 0.217 Ω.
3. All coils wound the same direction; reversal at connection (doc 18 Appendix A): 15 coils per phase in series, star.
4. Tests before assembly (doc 18 §5.7): resistance ±2 % of the lot mean, surge 3.5 kV, hipot to the segment 2.3 kV 1 min, IR ≥ 1 GΩ.
5. After assembly: VPI class H epoxy, end windings potted (doc 18 §5.11, §5.12). PT100 in 6 separator gaps (2 per phase).

SIDE VIEW OF COIL ON TOOTH (1:1, simplified)



Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

COIL ON SEGMENT (WOUND SEGMENT)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-203C	C · 2026-11-05	2:1 / 1:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
Cu enamelled Ø1.55 grade 2, class 200; NKN 0.30; PPS 4540 caps	≈ 0.45 kg Cu per coil	Claude / not checked